TEXORA · LECTURE SHEET

Automation in Textile Industry

Automation replaces watchfulness with measurement and response: a sensor detects, a controller decides, an actuator acts — in milliseconds, on every pick. This sheet studies four weaving-centred systems in depth: automatic warp tension control, automatic pick repair, electronic warp and weft stop motions, and online fabric fault monitoring.

TE-4105 · INDUSTRIAL UTILITIES · TOPIC 6 · LEVEL 4 TERM 1

1. What automation means on the loom floor

An automatic system is a **closed loop**: **sensor → controller → actuator → process → sensor again**. Open-loop aids (alarms, counters) inform the operator; closed-loop systems act themselves. The four systems of this sheet share that skeleton, and each exists because a human cannot watch hundreds of ends and picks continuously without missing the one that breaks. Automation gains are: fewer defects woven in, less loom downtime waiting for attention, consistent quality between shifts, and data (stops per shift, by cause) for maintenance and planning (Topic 9).

2. Automatic warp tension control

Problem. As the warp beam empties, its diameter falls; with a fixed let-off drive the warp tension would drift, changing pick spacing, fabric width and cover, and raising end breaks. Cloth fell and take-up must also stay in step.

System. A sensing roller or dancer rests on the warp sheet (or a load cell measures back-rest force). The controller compares measured tension with the set value and continuously adjusts the **let-off motor/servo** speed — releasing warp faster as the beam empties, or easing off as tension rises. Modern looms integrate this with electronic take-up, so warp tension and picks per cm are both held by recipe for each style.

Result. Constant tension from full beam to empty: uniform fabric, fewer tension bars and starting marks, fewer end breaks, and style change by setting rather than by trial.

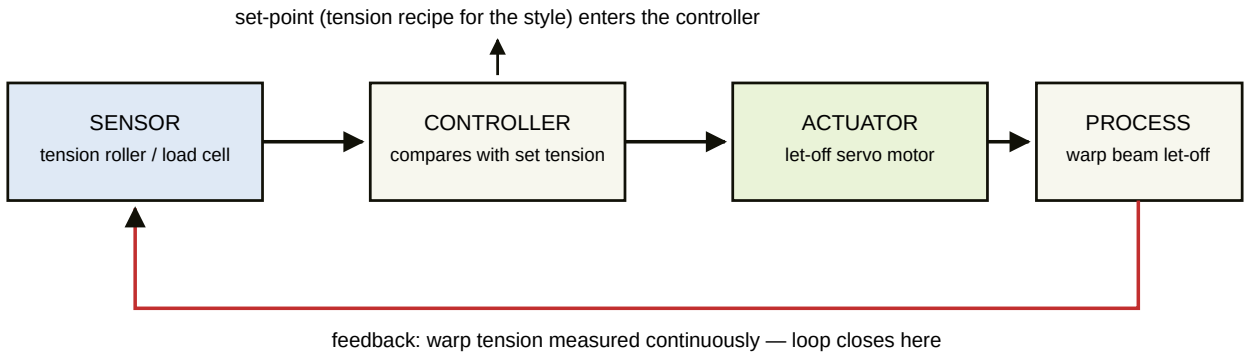


Figure 1 — Closed loop for automatic warp tension control. Signal flows sensor → controller → actuator → process; the red feedback path returns the measured tension to the sensor.

3. Automatic pick repair (pick finding and repair)

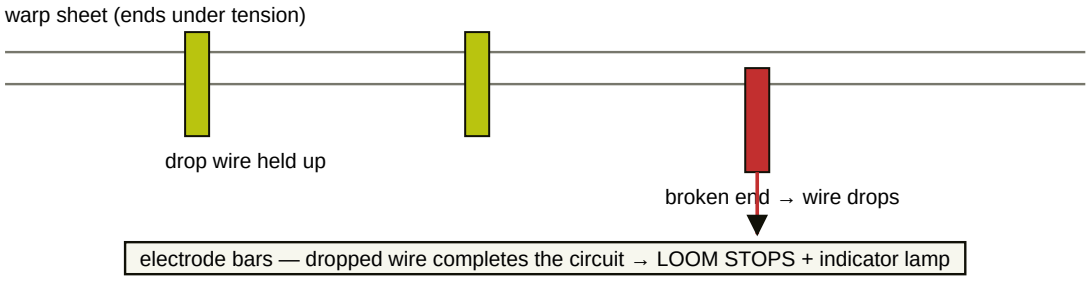
Problem. When a weft breaks or a pick is faulty, the loom stops with the broken pick partly woven. Manually, the weaver must find the pick, unweave to the fault and restart — slow, and a common source of starting marks.

System. On a stop for a weft fault, the loom automatically: (1) reverses the shedding/take-up to the **pick-finding** position, opening the shed containing the faulty pick; (2) removes or allows removal of the broken pick (some systems eject it automatically); (3) a fresh pick is inserted (on air-jet looms the insertion system re-fires after the fault is cleared); (4) the loom restarts in exact register, so no thick or thin place appears. Full automatic pick repair is a top-end feature, but pick finding alone removes most of the time and quality loss of a weft stop.

4. Electronic warp and weft stop motions

Warp stop motion. Each warp end passes through a **drop wire**. While the end is intact its tension holds the wire up; when the end breaks, the wire drops onto a pair of electrode bars, completing a circuit that stops the loom instantly and lights an indicator showing the zone of the break. Electronic versions sense the drop contactlessly and locate the broken end precisely. The older mechanical form uses the falling wire to shift a knock-off lever.

Weft stop motion. A sensor must confirm that every inserted pick actually arrived. Forms include: piezoelectric/optical sensors at the receiving side of air-jet and rapier looms that detect the weft tip each pick; fork-type feelers on shuttle looms. If the signal for a pick is missing, the loom stops *before* the next pick beats up the fault, and signals the cause so the weaver (or the pick-repair system, §3) can act.



Electronic warp stop motion: one drop wire per end; a break drops the wire and stops the loom within the pick.

Figure 2 — Electronic warp stop motion. Intact ends hold their drop wires up; the broken end's wire (red) falls onto the electrode bars, completing the stop circuit.

5. Online fabric fault monitoring

Problem. Traditional inspection happens after weaving, on a separate machine — faults are found only after metres of defective cloth exist, and loom faults (a damaged reed, a tension fault) keep producing them meanwhile.

System. Camera- or laser-based inspection heads mounted on the loom scan the moving cloth continuously. Image processing compares the fabric pattern with the learned good pattern and classifies deviations: missing or broken pick/end (already prevented by stop motions, but double-checked), oil spots, holes, slubs, colour or pattern faults, and edge defects. The system raises an alarm, can stop the loom for major faults, marks the fault position, and builds a **fabric map** per roll that follows the cloth to inspection and cutting — so cutting rooms plan around known faults instead of discovering them.

Result. Faults are caught at the pick they occur; defective length per fault falls from metres to centimetres; quality data by loom and shift guides maintenance.

SYSTEM	MACHINE	SENSOR	ACTION ON DETECTION
Warp tension control	Loom (let-off)	Tension roller / load cell	Controller adjusts let-off servo continuously
Automatic pick repair	Loom (shedding/take-up + insertion)	Weft-stop signal + position encoders	Pick finding, faulty pick removed, fresh pick, restart in register
Warp stop motion	Loom (warp line)	Drop wire per end + electrode bars	Loom stops; indicator shows break zone
Weft stop motion	Loom (receiving side)	Piezo / optical weft sensor	Loom stops before beat-up of the missing pick

Online fault monitoring	Loom (cloth line)	Camera / laser inspection head	Alarm, optional stop, fault mapped on the roll

6. Benefits, limits and the human role

- **Benefits:** higher loom efficiency (less waiting, faster repair), fewer woven-in defects, consistent quality across shifts, automatic production and stop-cause data.
- **Limits:** capital and maintenance cost; sensors foul with fly and size dust — cleaning and calibration are part of the maintenance plan; a mis-set system causes false stops that destroy the efficiency it was bought for.
- **Human role shifts, not vanishes:** from watching and reacting to setting, cleaning, analysing stop data and maintaining the systems themselves.

Glossary

Closed loop — a control system whose output is measured and fed back to correct itself. **Let-off** — controlled release of warp from the beam. **Pick finding** — automatic positioning of the loom at the shed containing a faulty pick. **Drop wire** — a wire held up by each warp end; it falls when the end breaks. **Stop motion** — a device that stops the machine when a fault (broken end/pick) occurs. **Online monitoring** — inspection carried out during production on the machine itself.

Key points

• Every system here is sensor → controller → actuator → process → feedback.
• Warp tension control holds tension constant as the beam empties by varying let-off servo speed.
• Stop motions stop the loom within the pick: drop wires for warp, arrival sensors for weft.
• Pick repair + online monitoring turn a fault from metres of defective cloth into a single corrected pick.

Practice

1. Draw the closed loop for weft stop motion, naming a real sensor and the actuator it triggers.
2. A loom's end breaks rise steadily as each beam nears its end, though the beam starts well. Name the failing system, the physical cause, and the fabric faults it produces.
3. Explain two ways online fabric monitoring saves money that final inspection cannot.
4. Why can a badly adjusted stop motion reduce loom efficiency instead of raising it?

Answer key

1. Sensor: piezo/optical weft-arrival sensor at the receiving side. Controller: loom control unit checking for the arrival signal each pick. Actuator: loom brake/drive cut (and signal lamp). Feedback: next pick's arrival confirms normal running.
2. Automatic warp tension control. As beam diameter falls, a fixed let-off gives rising tension variation/drift, raising breaks; faults: variable pick spacing, width variation, tension bars and starting marks.
3. (1) It stops fault production at the first defective pick, so defective length per incident falls sharply. (2) Its fault map lets cutting plan around known defects, raising cutting yield — final inspection finds faults only after a whole roll is woven.
4. False stops: oversensitive or fouled sensors stop the loom without a real fault; each stop costs production time and risks a starting mark, so net efficiency and quality fall.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib